Dear Search Committee,

In 2024, as a summer intern of the Data Science Summer Institute at Lawrence Livermore National Laboratory (LLNL), I worked on a project to understand how to use active learning and surrogate modeling to increase efficiency for computer experiments. Through this internship, I also worked with engineers to understand the problems they hope to use statistical models to solve. Further, I got to learn about the working environment and culture at a national laboratory by talking to several applied statisticians. After the internship concluded, I stayed as a collaborator to continue working on the project. We developed a new active learning algorithm, completed a manuscript, and am currently waiting for the completion of code review, after which we will submit the manuscript to a journal. Currently a PhD candidate, I have been contemplating the next step of my career. I am beyond excited to see the Applied Statistics Applied Statistics and Machine Learning Postdoc opening.

During my PhD study in statistics, I take full advantage of the wide range of faculty expertise offered by my department at NC State University and work on three research projects in different directions broadly connected by the concept centering uncertainty quantification (UQ). My first project, with Dr. Brian Reich, is about spatial and causal inference; we developed a fully Bayesian, spectral method to account for unmeasured spatial confounding and provide unbiased effect estimates and conservative credible intervals for multiple exposures and multiple outcomes. The project that started from the summer internship at LLNL, as mentioned above, became my second project. With Dr. Kevin Quinlan (LLNL) and Dr. Annie Booth (Virginia Tech), I developed an active learning method that utilizes Gaussian processes as surrogate models and joint predictive probabilities as acquisition functions to efficiently find sampling points to locate the optimal design for contour locations from multiple computer experiments. After two projects motivated by applications, I turned to fundamentals and am currently applying possibilistic inferential models (IM) to decision theory with Dr. Ryan Martin and Dr. Jonathan Williams. Using possibility measures, IM incorporates imprecision and provides validity guarantees. These projects provide breadth in thinking and wide opportunities for current and future collaboration. Importantly, even though IM has not been widely used in the statistics community, I believe it will greatly benefit research on physical process modeling, uncertainty quantification, and calibration.

I am especially drawn to LANL for the diverse opportunities within one organization. I am interested in collaborating with researchers outside statistics, mentoring future scholars, collaborating with statisticians in other academic institutes, and working on my own methodological research, all of which, as I understand, are possible at LANL. I hope to have the opportunity to learn more about this position and discuss what I can contribute to your team.

Sincerely,
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